ODOO ERP SYSTEM

Software Requirements Specification (SRS)

Enterprise Business Management Platform

Document attribute	Value
Document identifier	ODOO-ERP-SRS-001
Version	1.0 — Baseline draft
Status	For stakeholder review and approval
Date	11 October 2026
Scope	Integrated Odoo ERP business processes and supporting administration
Audience	Business owners, process leads, implementation, QA, IT and operations

This specification defines business-oriented functional requirements, quality requirements, business rules, data requirements, interfaces, access controls, and acceptance criteria for an Odoo-based ERP implementation. It follows a conventional enterprise SRS structure and is intended to serve as a controlled requirements baseline.

Document control: changes to approved requirements must be reviewed, versioned, and traceable.

Document Control

Version	Date	Change description	Approval
0.1	11 Oct 2026	Initial compilation of domain requirements	Pending
1.0	11 Oct 2026	Structured baseline for stakeholder review	Pending

Approval record

Role	Name	Decision	Date
Business sponsor	To be assigned	Pending	—
ERP product owner	To be assigned	Pending	—
Finance process owner	To be assigned	Pending	—
Operations / inventory owner	To be assigned	Pending	—
IT / security owner	To be assigned	Pending	—

“Shall” identifies a mandatory requirement. Requirements are uniquely identified so they can be reviewed, implemented, tested, and changed without depending on page numbers. Configurable values such as tax rules, approval limits, numbering, retention periods, and service targets require approval by the accountable business owner.

Table of Contents

Document Control . . . 2
Approval record . . . 2
Table of Contents . . . 3
1. Introduction . . . 9
1.1 Purpose . . . 9
1.2 Scope . . . 9
1.3 Business objectives . . . 9
1.4 Audience . . . 9
1.5 Conventions . . . 9
1.6 Basis and interpretation . . . 9
1.7 Product assumptions . . . 9
2. Overall Description . . . 9
2.1 Product perspective . . . 9
2.2 Capability groups . . . 10
2.3 User classes . . . 10
2.4 Operating environment . . . 10
2.5 Constraints . . . 10
2.6 Dependencies . . . 10
3. End-to-End Business Processes . . . 10
3.1 Lead-to-cash . . . 10
3.2 Procure-to-pay . . . 11
3.3 Inventory movement . . . 11
3.4 Plan-to-produce . . . 11
3.5 Record-to-report . . . 11
3.6 Hire-to-pay . . . 11

3.7 Case-to-resolution . . . 11

3.8 Plan-to-maintain . . . 11

3.9 Recurring service and rental . . . 11

3.10 Publish and manage content . . . 11

4. Functional Requirements — Common Services . . . 11

5. CRM . . . 15

Purpose and scope . . . 15

Principal business records . . . 15

Functional requirements . . . 15

6. Sales Management . . . 18

Purpose and scope . . . 18

Principal business records . . . 18

Functional requirements . . . 18

7. Purchase Management . . . 20

Purpose and scope . . . 20

Principal business records . . . 20

Functional requirements . . . 20

8. Inventory and Warehouse . . . 22

Purpose and scope . . . 22

Principal business records . . . 22

Functional requirements . . . 22

9. Manufacturing . . . 24

Purpose and scope . . . 24

Principal business records . . . 24

Functional requirements . . . 24

10. Accounting and Payments . . . 26

Purpose and scope . . . 26
Principal business records . . . 26
Functional requirements . . . 26
11. Point of Sale . . . 28
Purpose and scope . . . 28
Principal business records . . . 28
Functional requirements . . . 28
12. HR and Payroll . . . 30
Purpose and scope . . . 30
Principal business records . . . 30
Functional requirements . . . 30
13. Helpdesk and Customer Support . . . 32
Purpose and scope . . . 32
Principal business records . . . 32
Functional requirements . . . 32
14. Project Management . . . 34
Purpose and scope . . . 34
Principal business records . . . 34
Functional requirements . . . 34
15. Maintenance Management . . . 36
Purpose and scope . . . 36
Principal business records . . . 36
Functional requirements . . . 36
16. Subscriptions . . . 38
Purpose and scope . . . 38
Principal business records . . . 38

Functional requirements . . . 38

17. Rental Management . . . 40

Purpose and scope . . . 40

Principal business records . . . 40

Functional requirements . . . 40

18. Website and eCommerce . . . 42

Purpose and scope . . . 42

Principal business records . . . 42

Functional requirements . . . 42

19. Documents and Knowledge . . . 44

Purpose and scope . . . 44

Principal business records . . . 44

Functional requirements . . . 44

20. Quality Management . . . 46

Purpose and scope . . . 46

Principal business records . . . 46

Functional requirements . . . 46

21. Studio and Configuration Governance . . . 48

Purpose and scope . . . 48

Principal business records . . . 48

Functional requirements . . . 48

22. Business Rules . . . 50

23. Data Requirements . . . 51

23.1 Core master data . . . 51

23.2 Transaction data . . . 51

23.3 Data quality . . . 51

23.4 Retention and archive . . . 51
23.5 Migration . . . 51
23.6 Audit and provenance . . . 51
24. External Interface Requirements . . . 52
24.1 User interface . . . 52
24.2 APIs and integrations . . . 52
24.3 Hardware interfaces . . . 52
24.4 Error conventions . . . 52
25. Non-Functional Requirements . . . 53
25.1 Performance verification . . . 53
25.2 Recovery objectives . . . 53
25.3 Security verification . . . 54
26. Access Control and Separation of Duties . . . 54
26.1 Privileged access . . . 54
26.2 Separation of duties . . . 54
27. Reports, Dashboards and Monitoring . . . 55
27.1 Report controls . . . 55
28. Workflow, Validation and Exception Handling . . . 55
28.1 Validation principles . . . 55
28.2 Technical evidence . . . 56
28.3 Retry and recovery . . . 56
29. System Acceptance Criteria . . . 56
29.1 Acceptance process . . . 56
29.2 Exit criteria . . . 57
30. Traceability and Change Management . . . 57
30.1 Traceability . . . 57

30.2 Change request contents . . . 57

30.3 Baseline policy . . . 57

31. Assumptions, Open Decisions and Risks . . . 57

31.1 Decisions required before configuration . . . 57

31.2 Key risks . . . 58

31.3 Assumptions . . . 58

32. Glossary . . . 58

Appendix A. Suggested End-to-End Acceptance Scenarios . . . 59

Appendix B. Requirements Coverage Index . . . 60

Appendix C. Review Checklist . . . 60

1. Introduction

1.1 Purpose

This Software Requirements Specification establishes a common understanding of the expected behavior and quality of the Odoo ERP System among business stakeholders, implementation specialists, developers, testers, administrators, and support personnel. It describes what the system must do; detailed technical design and deployment decisions are maintained separately.

1.2 Scope

The system provides a unified business application for CRM, sales, purchasing, inventory, manufacturing, accounting, point of sale, HR, payroll, helpdesk, projects, maintenance, subscriptions, rental operations, eCommerce, website management, documents, knowledge content, quality management, and controlled application configuration. It supports consistent master data, traceable transactions, role-based access, approval controls, reporting, and auditable business activity.

1.3 Business objectives

- Maintain a consistent view of customers, suppliers, products, employees, locations, and transactions.
- Reduce duplicate entry by connecting operational and financial processes.
- Make transaction state, ownership, approvals, exceptions, and history visible to authorized users.
- Improve stock accuracy through controlled receipts, transfers, reservations, deliveries, counts, and returns.
- Support order-to-cash, procure-to-pay, production, employee administration, service delivery, and financial close.
- Provide reliable operational and financial reports for decision-making and compliance review.

1.4 Audience

The audience includes business process owners, department managers, ERP analysts, solution architects, end users, implementation partners, developers, quality assurance personnel, information-security reviewers, and support teams.

1.5 Conventions

- FR denotes functional requirements; NFR denotes non-functional requirements; BR denotes business rules; AC denotes acceptance criteria.
- Draft, confirmed, approved, posted, cancelled, closed, and archived represent distinct lifecycle states.
- “Shall” is mandatory, “should” is recommended, and “may” is optional.
- Country-specific tax, payroll, privacy, and statutory configuration must be confirmed for the selected deployment.

1.6 Basis and interpretation

The scope was consolidated from structured test-case, test-execution, and defect records (20,000, 20,000, and 10,000 rows), ten application log files (15,000 log records), and ten screen-capture/OCR batches (14,000 screen records). These materials indicate represented business areas and transaction vocabulary. Individual screen labels and narrative examples are not treated as approved business policy; requirements below express a coherent business baseline for review by the relevant process owners.

1.7 Product assumptions

The solution is Odoo-based. The exact Odoo version, edition, installed modules, localization, hosting model, integrations, and license constraints must be confirmed during planning. This document defines business capabilities rather than asserting that every capability is enabled in a specific installation.

2. Overall Description

2.1 Product perspective

The ERP is an integrated application in which business objects are linked across departments. A sales order may reserve stock, generate a delivery, and create an invoice; a purchase order may create an expected receipt and supplier bill; a manufacturing order may consume components and create finished goods; a support case may refer to a customer, product, order, delivery, or invoice. Links must preserve identity and transaction meaning across modules.

2.2 Capability groups

Group	Scope	Typical records
Commercial	CRM, Sales, Subscriptions, Rental, eCommerce, Website, POS	Lead, opportunity, quotation, order, contract, receipt
Supply chain	Purchase, Inventory, Manufacturing, Quality, Maintenance	Supplier, purchase order, transfer, lot/serial, BoM, work order
Finance	Accounting, invoices, payments, reconciliation, tax	Journal entry, invoice, credit note, payment, reconciliation
People	HR and Payroll	Employee, department, contract, payslip
Service and content	Project, Helpdesk, Knowledge, Documents	Project, task, ticket, article, attachment
Administration	Studio and configuration	Custom field, view, access group, workflow setting

2.3 User classes

User class	Responsibilities	Access principle
Operator	Create and process assigned operational records	Minimum rights for assigned duties
Manager	Review work queues, approve within authority, monitor exceptions	Team or department scope
Finance	Invoices, payments, journals, reconciliation, close	Finance-specific permissions and separation of duties
Warehouse / production	Receipts, transfers, work orders, checks	Location and operation scope
HR / payroll	Employee, contract and payroll processes	Restricted personal and compensation data
Auditor	Review records, approvals and reports	Read-only scope
Administrator	Users, configuration, integrations and recovery	Privileged access, monitored and controlled

2.4 Operating environment

The system shall operate in supported desktop and mobile browsers for the selected Odoo release. Supported browsers, device sizes, peripherals, deployment topology, database, file storage, identity provider, email, and integrations shall be documented in the deployment design. Concurrent operations shall preserve transaction consistency.

2.5 Constraints

- Prefer standard Odoo capabilities where they meet approved requirements; customizations require documented justification.
- Document custom models, fields, views, automations, integrations, and upgrade implications.
- Apply relevant accounting, tax, employment, privacy, and retention obligations.
- Protect restricted information in screens, reports, exports, notifications, APIs, and attachments.
- Failed operations must not leave partial stock, accounting, or workflow transactions.

2.6 Dependencies

- Approved master data and ownership rules for products, customers, suppliers, employees, units, locations, taxes, and accounts.
- Configured company, currency, fiscal year, journals, taxes, warehouses, routes, and stock operation types.
- Defined roles, approval limits, document numbering, email settings, and integration credentials.
- Availability and service terms of any selected payment, shipping, identity, banking, payroll, or eCommerce services.

3. End-to-End Business Processes

3.1 Lead-to-cash

Capture and qualify a prospect, manage an opportunity, prepare a quotation, obtain approval, confirm the order, fulfill goods/services, issue the invoice, register payment, and reconcile the account. Customer, order, delivery, invoice, and payment references remain connected.

3.2 Procure-to-pay

Prepare a purchase request/order, obtain approval, dispatch to supplier, receive goods/services, record discrepancies, validate the supplier bill, register payment, and reconcile payables. Receipt and bill quantities are compared under the approved matching policy.

3.3 Inventory movement

Receive, inspect, put away, reserve, pick, pack, ship, transfer, count, adjust, or return stock. Each movement records product, quantity, unit, source, destination, date, responsible user, and linked document as applicable.

3.4 Plan-to-produce

Maintain bills of materials and routing data, plan production, check component availability, release a manufacturing order, consume components, record operations and quality checks, produce finished goods, and close with traceable quantities and variances.

3.5 Record-to-report

Capture financial events in journals, post invoices and payments, reconcile open items, review financial statements, manage period controls, and close accounting periods under authorized supervision.

3.6 Hire-to-pay

Maintain employee and employment records, process approved time/work inputs where enabled, prepare payslips, review payroll totals, confirm payroll, and record resulting accounting entries.

3.7 Case-to-resolution

Register a support request, categorize and assign it, assess priority and service target, communicate with the requester, link relevant records, document resolution, and close or reopen with history preserved.

3.8 Plan-to-maintain

Register equipment and maintenance plans, create preventive or corrective requests, assign technicians, record work and parts, verify completion, and preserve equipment history and downtime.

3.9 Recurring service and rental

Create a subscription or rental agreement, define dates and commercial terms, manage renewals or returns, calculate charges using approved rules, invoice the customer, and handle changes, cancellation, overdue items, and disputes.

3.10 Publish and manage content

Create website/eCommerce content, maintain product and availability information, manage documents and knowledge articles, control publication state, and preserve access restrictions and revision history.

4. Functional Requirements — Common Services

FR-COM-001 — Authentication

Requirement: The system shall require users to authenticate before accessing protected business records.

Rationale: Actions must be attributable to an identified user.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-002 — Role-based authorization

Requirement: The system shall separately authorize read, create, edit, approve, post, export, delete, and administer actions by role and record scope.

Rationale: Different duties and sensitive data require differentiated permissions.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-003 — Company context

Requirement: The system shall enforce active company and permitted company scope when displaying or processing records.

Rationale: Prevents unintended cross-company access and accounting contamination.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-004 — Lifecycle controls

Requirement: The system shall define valid states and transitions for each supported record type.

Rationale: Incomplete or unapproved transactions must not be treated as final.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-005 — Audit history

Requirement: The system shall retain creator, timestamps, material changes, state changes, approvals, and posting history for configured high-impact records.

Rationale: Supports accountability, support, and compliance review.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-006 — Search and filtering

Requirement: The system shall provide appropriate list, search, filter, sort, grouping, and saved-filter capabilities.

Rationale: Users need efficient access to records within their permissions.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-007 — Notes and attachments

Requirement: The system shall associate authorized notes, messages, and attachments with supported records while enforcing parent-record access.

Rationale: Evidence and communication should remain in context.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-008 — Activities and notifications

Requirement: The system shall support assigned activities, reminders, and notifications for defined events and due dates.

Rationale: Accountable users need visible work queues.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-009 — Import and export

Requirement: The system shall validate imported records before commit and restrict exports according to permissions and sensitivity.

Rationale: Bulk operations must not bypass controls.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-010 — Duplicate prevention

Requirement: The system shall enforce unique identifiers and configured duplicate checks for key master and transactional records.

Rationale: Duplicate business records create operational and financial risk.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-011 — Error handling

Requirement: The system shall display actionable, non-sensitive errors and retain safe diagnostics for authorized support personnel.

Rationale: Users need corrective guidance without exposure of technical internals.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-012 — Transaction integrity

Requirement: The system shall commit related changes atomically when a business operation requires an all-or-nothing outcome.

Rationale: Prevents partial stock, financial, or workflow updates.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-013 — Localization

Requirement: The system shall support configured language, date, time, number, currency, unit, and time-zone presentation.

Rationale: Consistent values are required across regions.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-014 — Configuration governance

Requirement: The system shall restrict production-impacting configuration to authorized administrators and record significant changes.

Rationale: Configuration affects many business processes.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

FR-COM-015 — Archiving

Requirement: The system shall archive eligible records without breaking historical references or reports.

Rationale: Historical evidence must remain referentially consistent.

Preconditions: An authenticated user has the required role and company scope.

Main flow: Open the record; enter or select validated data; review calculated values and warnings; save or confirm; verify linked history.

Exception handling: Invalid values, unauthorized actions, duplicate submissions, missing configuration, and invalid state transitions shall be rejected without leaving a partial transaction.

Acceptance evidence: Role tests, validation tests, audit review, and duplicate/retry tests demonstrate the control.

5. CRM

Purpose and scope

Manage prospects and sales opportunities before and during commercial engagement.

Principal business records

Lead, opportunity, contact, team, activity, expected revenue, probability, source, lost reason, quotation reference.

Functional requirements

FR-CRM-001 — Lead registration

Requirement: Create a lead with contact details, source, owner, team, status, and next action.

Rationale: Consistent prospect records reduce scattered information.

Preconditions: Required contact fields and duplicate rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Save a valid lead; missing fields are identified; configured duplicate warning appears.

FR-CRM-002 — Qualification and conversion

Requirement: Support qualification, conversion, disqualification, and assignment using configured criteria.

Rationale: A controlled transition turns inquiries into actionable opportunities.

Preconditions: Only permitted states may be selected; disqualification reason may be required.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Decision, user, time, and resulting linked records are reviewable.

FR-CRM-003 — Opportunity pipeline

Requirement: Manage stages with owner, expected revenue, probability, closing date, and next activity.

Rationale: Pipeline stages support forecast and follow-up.

Preconditions: Stage transitions follow configured requirements.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Opportunity appears in correct stage and contributes to filtered pipeline totals.

FR-CRM-004 — Activities

Requirement: Schedule calls, meetings, tasks, and reminders against leads, opportunities, and customers.

Rationale: Follow-up commitments must be visible.

Preconditions: Overdue activities remain visible until completed or cancelled.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Assigned, upcoming, and overdue activities can be reviewed.

FR-CRM-005 — Opportunity-to-quotation

Requirement: Link a qualified opportunity to the relevant customer and quotation without losing history.

Rationale: Commercial context should persist through sales.

Preconditions: Existing customers should be reused when a match is confirmed.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Quotation retains source opportunity reference.

FR-CRM-006 — Pipeline reporting

Requirement: Report opportunity counts/value by stage, owner, team, source, and expected closing period.

Rationale: Managers need sales progress and forecast visibility.

Preconditions: Currency conversion and access filters are applied consistently.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Totals reconcile to the included opportunity set.

6. Sales Management

Purpose and scope

Manage quotations, orders, pricing, customer commitments, fulfillment, invoices, returns, and order status.

Principal business records

Quotation, sales order, order line, customer, product, price list, tax, delivery, invoice, payment terms.

Functional requirements

FR-SAL-001 — Quotation creation

Requirement: Create quotations with customer, product/service lines, quantities, units, prices, discounts, taxes, delivery terms, and validity date.

Rationale: A quotation is the formal commercial offer.

Preconditions: Customer, currency, and line details must be valid.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Displayed totals match configured pricing and tax rules.

FR-SAL-002 — Pricing and discounts

Requirement: Calculate prices from configured price lists and apply authorized discounts and taxes.

Rationale: Pricing must be consistent and governed.

Preconditions: Discount thresholds and approval rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Unauthorized discounts are blocked or routed for approval.

FR-SAL-003 — Approval and confirmation

Requirement: Route quotations through configured approval rules before confirmation.

Rationale: Customer commitments must be authorized.

Preconditions: Only valid quotations in confirmable states may become orders.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Confirmation records user/time and prevents duplicate order creation.

FR-SAL-004 — Order amendment

Requirement: Control changes to confirmed orders and retain audit history of material changes.

Rationale: Changes may affect stock, delivery, and revenue.

Preconditions: Changes after delivery/invoicing follow correction workflows.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Prohibited edits are rejected; permitted amendment path is available.

FR-SAL-005 — Delivery linkage

Requirement: Create or link delivery operations from confirmed orders according to product and route configuration.

Rationale: Sales commitments connect to warehouse execution.

Preconditions: Availability and reservation policies govern delivery.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Order-to-delivery navigation shows delivered, remaining, and cancelled quantities.

FR-SAL-006 — Invoice, return and reporting

Requirement: Create invoices from eligible order lines and support traceable returns/credit notes and sales reporting.

Rationale: Billing and corrections must reflect actual business outcomes.

Preconditions: Quantity and refund rules follow the configured policy.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Invoice/credit lines retain source references and quantities reconcile.

7. Purchase Management

Purpose and scope

Manage supplier records, purchase requests/orders, approvals, receipts, supplier bills, returns, and purchasing analysis.

Principal business records

Supplier, purchase request, supplier quotation, purchase order, receipt, supplier bill, payment term.

Functional requirements

FR-PUR-001 — Supplier master

Requirement: Maintain supplier identity, contacts, addresses, payment terms, currency, and tax details.

Rationale: Supplier accuracy supports orders and payables.

Preconditions: Duplicate and required-field checks are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Blocked suppliers cannot be used for new orders unless policy permits.

FR-PUR-002 — Purchase order

Requirement: Create an order with supplier, items, quantity, unit price, expected date, taxes, and destination.

Rationale: Orders establish supplier commitments.

Preconditions: Supplier, product, unit, and price validations apply.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Totals and expected receipt lines are calculated consistently.

FR-PUR-003 — Approval

Requirement: Route purchase orders through configured spending thresholds before confirmation/dispatch.

Rationale: Spending authority must be controlled.

Preconditions: Approver groups and limits are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: A user lacking authority cannot finalize an order above their limit.

FR-PUR-004 — Receipt matching

Requirement: Link received quantities to orders and record partial receipts, shortages, overages, or damaged goods.

Rationale: Receipt evidence must match physical goods.

Preconditions: Tolerance and inspection rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Ordered, received, accepted, rejected, and remaining quantities are distinct.

FR-PUR-005 — Supplier bill matching

Requirement: Match supplier bills against orders and receipts and flag configured variances.

Rationale: Reduces duplicate or unsupported charges.

Preconditions: Material quantity/price mismatches are held or flagged.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Bill links to source records and mismatch handling is recorded.

FR-PUR-006 — Returns and reporting

Requirement: Support supplier returns and report open orders, overdue receipts, bills awaiting match, and commitments.

Rationale: Rejected goods and purchasing commitments need traceability.

Preconditions: Return authorization and valuation rules apply.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Return stock movement and report totals are traceable.

8. Inventory and Warehouse

Purpose and scope

Manage products, warehouses, locations, receipts, transfers, reservations, deliveries, lot/serials, replenishment, counts, and valuation integration.

Principal business records

Product, warehouse, location, stock move, picking, lot/serial, package, route, reorder rule, adjustment.

Functional requirements

FR-INV-001 — Product stock settings

Requirement: Maintain product type, stock tracking, units, routes, and logistics attributes.

Rationale: Inventory behavior depends on correct setup.

Preconditions: Only authorized users change stock-impacting settings.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Incompatible settings are rejected or existing-transaction impacts are shown.

FR-INV-002 — Goods receipt

Requirement: Record incoming goods against an order or authorized receipt with quantity, destination, tracking ID, and inspection state.

Rationale: Physical receipts must become controlled stock.

Preconditions: Tracked products require lot/serial details.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Validated receipt updates on-hand stock and retains source references.

FR-INV-003 — Internal transfer

Requirement: Move stock between permitted locations with source, destination, quantity, user, and state.

Rationale: Internal movement must preserve location accuracy.

Preconditions: Source stock and destination rules are validated.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Completed transfer updates source and destination balances consistently.

FR-INV-004 — Reservation and picking

Requirement: Reserve eligible stock for demand and support authorized picking operations.

Rationale: Fulfillment depends on correct allocation.

Preconditions: On-hand, reserved, forecast, and available quantities are distinct.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Shortages and affected demand are visible.

FR-INV-005 — Delivery and traceability

Requirement: Validate outbound shipments and retain actual quantities and lot/serial details; support forward/backward traceability.

Rationale: Delivery and traceability support customer service and recalls.

Preconditions: Only permitted quantities and states may be validated.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: A lot/serial can be traced through receipts, transfers, production, and delivery.

FR-INV-006 — Counts, replenishment and valuation

Requirement: Support physical counts, approved adjustments, replenishment suggestions, and configured valuation integration.

Rationale: Stock accuracy and availability affect operations and finance.

Preconditions: Material adjustments may require approval; valuation accounts must be configured.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Count variances are recorded; stock and valuation entries are traceable.

9. Manufacturing

Purpose and scope

Manage bills of materials, production orders, work centers, work orders, component consumption, output, scrap, and quality checks.

Principal business records

BoM, manufacturing order, work center, routing, work order, component move, finished product, scrap, lot.

Functional requirements

FR-MFG-001 — Bill of materials

Requirement: Maintain approved BoMs with finished product, components, quantities, units, and applicable variants.

Rationale: Production requires a controlled material definition.

Preconditions: Only approved BoMs may be used for new production.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Expected component demand is calculated from selected BoM and order quantity.

FR-MFG-002 — Manufacturing order

Requirement: Create an order with product, quantity, BoM, dates, source demand, and state.

Rationale: The order coordinates production execution.

Preconditions: Required component/routing setup must be valid.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Order shows expected components and current production state.

FR-MFG-003 — Component availability

Requirement: Show component availability and shortages before/during release.

Rationale: Production plans must account for material constraints.

Preconditions: Configured reservation and warehouse rules apply.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Shortages remain visible and are not shown as available stock.

FR-MFG-004 — Work execution

Requirement: Support work assignment, start/stop, operation completion, consumed quantities, and duration where enabled.

Rationale: Operations require traceable execution.

Preconditions: Only permitted users complete controlled operations.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Work history records operator, state changes, quantities, and timestamps.

FR-MFG-005 — Consumption, output and scrap

Requirement: Record component consumption, good output, scrap, and approved rework against the order.

Rationale: Production stock must reflect actual results.

Preconditions: Over-consumption and scrap reasons follow policy.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Consumed, produced, and scrapped quantities reconcile to the order.

FR-MFG-006 — Quality and close

Requirement: Apply configured quality checks and close completed orders with planned-versus-actual analysis.

Rationale: Quality gates and production analysis protect output.

Preconditions: Failed checks or incomplete operations may block closure.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Closed order retains inspection outcomes and variance summary.

10. Accounting and Payments

Purpose and scope

Manage accounting configuration, invoices, supplier bills, journals, payments, reconciliation, taxes, period controls, and financial reporting.

Principal business records

Chart of accounts, journal, invoice, bill, credit note, payment, bank statement, tax, fiscal position.

Functional requirements

FR-ACC-001 — Accounting configuration

Requirement: Support approved chart of accounts, journals, periods, currencies, taxes, payment terms, and localization.

Rationale: Financial transactions require consistent configuration.

Preconditions: Posting is blocked when required settings are missing.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Authorized reviewers verify configuration with test transactions.

FR-ACC-002 — Customer invoices and supplier bills

Requirement: Create invoices/bills from eligible transactions and support approved manual entry.

Rationale: Invoices formalize receivables and payables.

Preconditions: Required parties, dates, lines, taxes, and journals must validate.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Totals, due dates, taxes, and source references are visible.

FR-ACC-003 — Posting controls

Requirement: Separate draft from posted documents and restrict posting to authorized users.

Rationale: Posted entries affect official balances.

Preconditions: Corrections use approved reversal/credit/debit paths.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Posting user/time and journal sequence are recorded.

FR-ACC-004 — Payments and allocation

Requirement: Register receipts/payments and allocate them to eligible invoices/bills.

Rationale: Cash movement must reduce correct balances.

Preconditions: Payment method, currency, journal, and allocation are validated.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Open, partial, paid, and overpaid positions are represented correctly.

FR-ACC-005 — Reconciliation and period lock

Requirement: Match bank/payment lines to accounting entries and restrict postings into locked periods.

Rationale: Reconciliation and period control support close.

Preconditions: Overrides require authorization and an audit record if allowed.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Matched and unmatched items are distinguishable; unauthorized backdating is rejected.

FR-ACC-006 — Tax and financial reports

Requirement: Apply configured tax rules and provide general ledger, trial balance, financial statements, and aging reports.

Rationale: Finance needs accurate statutory and management reporting.

Preconditions: Tax settings are maintained by authorized finance roles.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Report totals reconcile to posted entries under approved scenarios.

11. Point of Sale

Purpose and scope

Support retail/counter sales, product selection, pricing, payments, receipts, returns, cash movements, and session reconciliation.

Principal business records

POS configuration, session, order, order line, customer, payment, receipt, refund, cash movement.

Functional requirements

FR-POS-001 — Session opening

Requirement: Open a session with configured POS, responsible cashier, and opening balance where applicable.

Rationale: Sessions establish accountability.

Preconditions: Concurrent session policy is configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Opening user, time, and opening balance are recorded.

FR-POS-002 — Sale creation

Requirement: Select products, quantities, customer, prices, taxes, and permitted discounts.

Rationale: POS must capture transactions accurately.

Preconditions: Active product and pricing rules apply.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Order total and tax breakdown are calculated correctly.

FR-POS-003 — Payment completion

Requirement: Support configured payment methods and mark an order paid only when conditions are met.

Rationale: Order state must reflect payment evidence.

Preconditions: Split payments, change, and rounding follow configuration.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Paid order contains payment lines and a receipt reference.

FR-POS-004 — Receipt and refund

Requirement: Generate receipts and support traceable refunds linked to original transactions where possible.

Rationale: Customers need evidence and refunds need traceability.

Preconditions: Refund quantities and approvals follow policy.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Refund records identify original lines and stock/finance effects.

FR-POS-005 — Session close

Requirement: Calculate expected totals by payment method and support cashier reconciliation.

Rationale: Session close reconciles sales and collections.

Preconditions: Differences are recorded and may require approval.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Expected, counted, and variance amounts are shown.

FR-POS-006 — POS reporting

Requirement: Report sales by session, cashier, product, payment method, and returns.

Rationale: Managers need daily sales visibility.

Preconditions: Reports follow session, date, and company filters.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Totals reconcile to the selected POS orders and payments.

12. HR and Payroll

Purpose and scope

Maintain employee and employment records; calculate and account for compensation where payroll functionality is enabled.

Principal business records

Employee, department, job, manager, contract, work inputs, salary structure, payslip, payroll batch.

Functional requirements

FR-HR-001 — Employee master

Requirement: Maintain employee identity, work contact, department, position, manager, and employment status.

Rationale: Reliable employee data supports workforce administration.

Preconditions: Access to personal fields is limited to authorized HR roles.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Activation requires configured mandatory information.

FR-HR-002 — Employment lifecycle and contracts

Requirement: Track effective-dated changes, contract dates, working arrangements, and current employment state.

Rationale: Employment changes require controlled history.

Preconditions: Overlapping/expired contracts are highlighted according to policy.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Current state and effective-dated history can be reviewed.

FR-HR-003 — Personnel confidentiality

Requirement: Restrict personal, contractual, compensation, and sensitive HR documents to authorized roles.

Rationale: Employee data is confidential.

Preconditions: Exports, attachments, reports, and APIs enforce equivalent controls.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Unauthorized role cannot retrieve restricted data through alternate paths.

FR-PAY-001 — Payroll inputs

Requirement: Collect/import approved payroll inputs and identify missing or inconsistent data before calculation.

Rationale: Payroll accuracy depends on controlled inputs.

Preconditions: Only approved inputs are included.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Run summary identifies included inputs and validation issues.

FR-PAY-002 — Payslip calculation

Requirement: Calculate payslips using approved salary structure, contract, period, and configured rules.

Rationale: Compensation must be consistent.

Preconditions: Localization and calculation rules are reviewed before production.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Representative payroll scenarios match approved expected results.

FR-PAY-003 — Payroll confirmation and accounting

Requirement: Support draft, review, correction, confirmation, and traceable accounting entries where enabled.

Rationale: Payroll is a high-impact financial process.

Preconditions: Only authorized users confirm batches.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Confirmed totals reconcile to employee count and accounting entries.

13. Helpdesk and Customer Support

Purpose and scope

Manage service requests from intake through classification, assignment, response, resolution, and closure.

Principal business records

Ticket, requester, customer, team, assignee, priority, category, SLA target, message, related record.

Functional requirements

FR-HD-001 — Ticket intake

Requirement: Create a ticket with requester, subject, description, category, priority, and received time.

Rationale: Support work needs a trackable record.

Preconditions: Required fields and intake channels are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Valid ticket receives a unique reference and initial state.

FR-HD-002 — Assignment and routing

Requirement: Assign tickets to teams/users using configured rules and preserve reassignment history.

Rationale: Ownership reduces unassigned work.

Preconditions: Only permitted roles may reassign.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Current owner and assignment history are visible.

FR-HD-003 — Priority and service targets

Requirement: Calculate/display response and resolution targets where service rules are enabled.

Rationale: Urgency and commitments must be visible.

Preconditions: Pause/wait/reopen timing follows configured rules.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Overdue tickets are identifiable in lists and reports.

FR-HD-004 — Communication history

Requirement: Associate messages, internal notes, attachments, and status changes with the ticket.

Rationale: Support context must be preserved.

Preconditions: Internal notes shall not be sent as customer messages.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Roles see only communication types they are authorized to view.

FR-HD-005 — Related records

Requirement: Link permitted customer, product, order, delivery, or invoice records to tickets.

Rationale: Resolution depends on business context.

Preconditions: Linked records respect access restrictions.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Agents can navigate to permitted records without losing ticket context.

FR-HD-006 — Resolution and reporting

Requirement: Capture resolution, close/reopen with history, and report aging, workload, and service measures.

Rationale: Repeated issues and service quality need visibility.

Preconditions: Resolution fields and business calendars are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Closed tickets retain timeline; reports reconcile to ticket records.

14. Project Management

Purpose and scope

Organize work into projects, tasks, milestones, dependencies, owners, deadlines, and effort tracking where enabled.

Principal business records

Project, task, milestone, stage, assignee, dependency, timesheet, customer, analytic account.

Functional requirements

FR-PRJ-001 — Project setup

Requirement: Create projects with owner, status, dates, customer/internal designation, and visibility.

Rationale: Projects need clear ownership and scope.

Preconditions: Access and visibility are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Authorized users can find project and current state.

FR-PRJ-002 — Task lifecycle

Requirement: Manage tasks with description, assignee, stage, priority, deadline, and linked project.

Rationale: Tasks represent actionable work.

Preconditions: Transitions may require completion details.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Board and list views show consistent ownership and state.

FR-PRJ-003 — Milestones and dependencies

Requirement: Support milestones and dependencies where enabled and show blocked/overdue work.

Rationale: Dependencies affect delivery.

Preconditions: Circular dependencies are rejected where supported.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Blocked work is distinguishable from ready work.

FR-PRJ-004 — Time tracking

Requirement: Where enabled, record work date, employee, task, duration, and approval state.

Rationale: Effort supports costing and billing.

Preconditions: Approval and lock rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Approved time appears in authorized project summaries.

FR-PRJ-005 — Project reporting

Requirement: Report task status, overdue work, workload, milestones, and progress.

Rationale: Managers need delivery visibility.

Preconditions: Reports respect project access rights.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Totals and overdue indicators match filtered task records.

15. Maintenance Management

Purpose and scope

Manage equipment, preventive schedules, corrective requests, technician assignments, work history, downtime, and spare-parts use.

Principal business records

Equipment, request, request type, preventive plan, technician, work center, spare part, downtime event.

Functional requirements

FR-MNT-001 — Equipment register

Requirement: Maintain unique equipment ID, category, location, responsible team, and relevant warranty/supplier details.

Rationale: Maintenance needs identifiable assets.

Preconditions: Identifiers and mandatory fields are configured.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Equipment history can be reviewed.

FR-MNT-002 — Corrective request

Requirement: Report a fault with equipment, description, priority, date, and reporter.

Rationale: Breakdowns require triage.

Preconditions: Required details vary by request type.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Request receives a unique reference and owner.

FR-MNT-003 — Preventive schedule

Requirement: Support recurring maintenance by time or usage criteria where enabled.

Rationale: Planned work reduces avoidable failures.

Preconditions: Frequency and due-date calculations are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Due and overdue work is visible.

FR-MNT-004 — Work completion

Requirement: Record technician, work performed, findings, duration, and completion state.

Rationale: Equipment history supports reliability.

Preconditions: Only authorized users complete controlled requests.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Completed work is linked to equipment and retains summary.

FR-MNT-005 — Parts and downtime

Requirement: Associate spare parts and downtime with maintenance requests where configured.

Rationale: Maintenance affects stock and availability.

Preconditions: Parts use controlled stock movements.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Parts and downtime can be reported by equipment/request.

FR-MNT-006 — Maintenance reporting

Requirement: Report backlog, due work, downtime, and maintenance history.

Rationale: Leaders need workload and reliability insight.

Preconditions: Planned and corrective work are distinguishable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Reports reconcile to requests and recorded downtime.

16. Subscriptions

Purpose and scope

Manage recurring agreements, billing cadence, renewal, plan changes, suspension, cancellation, and customer communications.

Principal business records

Subscription, customer, plan, recurring line, start/end date, renewal, invoice, payment state, cancellation reason.

Functional requirements

FR-SUB-001 — Subscription creation

Requirement: Create a customer-linked subscription with plan, recurring lines, dates, cadence, currency, and terms.

Rationale: Recurring services require explicit contract details.

Preconditions: Plan and billing settings must be valid.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Next expected billing or renewal date is visible.

FR-SUB-002 — Recurring invoice

Requirement: Generate eligible recurring invoices according to schedule and billing policy.

Rationale: Recurring charges must be predictable.

Preconditions: Duplicate billing for the same period is prevented.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Invoice links to subscription and billing period.

FR-SUB-003 — Renewal management

Requirement: Identify upcoming renewals and support authorized renewal, change, or non-renewal decisions.

Rationale: Renewal affects customer commitments.

Preconditions: Reminders and approvals are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Upcoming and overdue renewals appear in work queues.

FR-SUB-004 — Plan change and proration

Requirement: Apply approved plan/quantity changes using configured effective-date and proration rules where enabled.

Rationale: Mid-cycle changes can affect charges.

Preconditions: Proration must be explicitly configured.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Invoice calculation is reproducible from approved change.

FR-SUB-005 — Suspension and cancellation

Requirement: Support suspension/cancellation with effective date, reason, and billing consequences.

Rationale: Service and billing status must align.

Preconditions: Permissions and refund rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Lifecycle history records decision and effective date.

FR-SUB-006 — Subscription reporting

Requirement: Report active, renewing, overdue, suspended, cancelled, and recurring amounts.

Rationale: Commercial teams need visibility.

Preconditions: Amounts identify currency and time basis.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Totals match selected subscription records.

17. Rental Management

Purpose and scope

Manage reservable items, rental periods, availability, pickup, return, inspection, late return, damage, deposits, and invoicing.

Principal business records

Rental product, customer, rental order, reservation period, pickup, return, inspection, deposit, damage charge.

Functional requirements

FR-RNT-001 — Rental order

Requirement: Create rental order with customer, items, quantities, dates/times, rates, and terms.

Rationale: Rental commitments depend on exact periods.

Preconditions: Date conflicts and unavailable items are validated.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Order shows reservation period and commercial total.

FR-RNT-002 — Availability

Requirement: Check availability for requested period according to stock and reservation rules.

Rationale: Prevents double-booking.

Preconditions: Overlaps are prevented or flagged for authorized resolution.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Incompatible reservations cannot both be confirmed.

FR-RNT-003 — Pickup and handover

Requirement: Record handover time, quantities, condition, deposit, and acknowledgment where required.

Rationale: Custody transfer should be traceable.

Preconditions: Condition fields are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Actual handed-over items are shown.

FR-RNT-004 — Return and inspection

Requirement: Record return time, quantities, condition, missing items, and inspection outcome.

Rationale: Return state determines availability and charges.

Preconditions: Damaged items require controlled disposition.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Items become available only after required checks.

FR-RNT-005 — Late/damage charges

Requirement: Calculate approved late/damage charges and manage deposits/refunds according to policy.

Rationale: Adjustments must be explainable.

Preconditions: Rates, grace periods, and approval thresholds are configured.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Invoice/refund shows reason and calculation basis.

FR-RNT-006 — Rental reporting

Requirement: Report utilization, current rentals, upcoming/overdue returns, damage, and revenue.

Rationale: Operations need availability visibility.

Preconditions: Time zone and period rules are consistent.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Overdue items trace to open rental orders.

18. Website and eCommerce

Purpose and scope

Publish business content, maintain product catalogs, support online ordering, and connect sales, inventory, and payment processes.

Principal business records

Website, page, menu, product listing, category, cart, online order, payment transaction, customer.

Functional requirements

FR-WEB-001 — Content publication

Requirement: Draft, preview, publish, unpublish, and update pages according to assigned rights.

Rationale: Public content must be controlled.

Preconditions: Only authorized roles publish production content.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Draft content is not public before publication.

FR-WEB-002 — Product catalog

Requirement: Display eligible products with approved descriptions, images, variants, prices, and availability.

Rationale: Online product information must be consistent.

Preconditions: Only publishable products and permitted prices are exposed.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Displayed product and price match catalog settings.

FR-WEB-003 — Cart and checkout

Requirement: Select products, update quantities, enter required details, and submit an order.

Rationale: Checkout creates a traceable sale.

Preconditions: Price, stock, tax, delivery and fields are revalidated.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: One valid checkout creates one order reference.

FR-WEB-004 — Payment outcome

Requirement: Where enabled, record payment attempts and distinguish successful, pending, failed, and cancelled states.

Rationale: Order state must reflect payment evidence.

Preconditions: Retries cannot duplicate confirmed payments/orders.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Provider reference and final outcome are visible to authorized staff.

FR-WEB-005 — Order integration

Requirement: Link online orders to sales and fulfillment and reflect relevant state changes.

Rationale: Online and back-office records must agree.

Preconditions: Reservation and cancellation rules apply consistently.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Staff can navigate to fulfillment records.

FR-WEB-006 — Public privacy

Requirement: Expose only intended public content and protect customer and operational data.

Rationale: Public features increase exposure risk.

Preconditions: Internal notes and restricted attachments are never public.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Security tests confirm private data is inaccessible without permission.

19. Documents and Knowledge

Purpose and scope

Manage business files, folders, tags, revisions, ownership, access, and reusable internal articles.

Principal business records

Document, attachment, folder/workspace, tag, owner, revision, article, publication state, linked record.

Functional requirements

FR-DOC-001 — Document registration

Requirement: Store/reference documents with title, type, owner, date, tags, and linked business record.

Rationale: Documents need discoverable context.

Preconditions: Metadata and file-type rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Authorized users can find by title, tag, owner, or link.

FR-DOC-002 — Document permissions

Requirement: Apply access permissions based on role, workspace, and parent-record rules.

Rationale: Documents may contain confidential data.

Preconditions: View, download, share, and export all check permission.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Direct-link access is denied to unauthorized users.

FR-DOC-003 — Revision and archive

Requirement: Preserve approved revisions and record material changes where enabled.

Rationale: Users must identify applicable version.

Preconditions: Retention and archive rules are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Current and prior retained versions are distinguishable.

FR-DOC-004 — Knowledge lifecycle

Requirement: Support draft, review, publish, update, and retire states for articles.

Rationale: Operational guidance must be controlled.

Preconditions: Publication may require reviewer approval.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Only published articles are shown to intended audience.

FR-DOC-005 — Search and linking

Requirement: Search authorized articles and link guidance to tickets/business records.

Rationale: Reusable knowledge improves support consistency.

Preconditions: Search results respect visibility rules.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Article and linked context can be opened by authorized user.

FR-DOC-006 — Retention and deletion

Requirement: Apply approved retention and deletion while preserving required records and references.

Rationale: Information lifecycle must be governed.

Preconditions: Legal holds override routine deletion.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Archive/delete events are recorded according to policy.

20. Quality Management

Purpose and scope

Define inspections, checks, pass/fail outcomes, nonconformance, corrective action, and traceability across receipts, production, and delivery.

Principal business records

Quality plan, inspection, alert, nonconformance, product/lot, operation, disposition, corrective action.

Functional requirements

FR-QLT-001 — Quality plan

Requirement: Define checks by product, operation, receipt, production stage, or delivery point.

Rationale: Quality controls must be repeatable.

Preconditions: Plan activation and versioning are controlled.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Representative operation triggers intended checks.

FR-QLT-002 — Inspection recording

Requirement: Record date, inspector, item/lot, criteria, result, and disposition.

Rationale: Inspection is evidence of conformance.

Preconditions: Required results are complete before controlled operation proceeds.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Inspection remains linked to source operation and product.

FR-QLT-003 — Nonconformance

Requirement: Create alert for failed checks with severity, owner, containment, and disposition.

Rationale: Failures require accountable resolution.

Preconditions: Disposition and escalation are configurable.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Failed items remain distinguishable until disposition.

FR-QLT-004 — Corrective action

Requirement: Record investigation, cause, action, owner, due date, and verification where enabled.

Rationale: Repeated failures need follow-through.

Preconditions: Closure may require evidence and verification.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Closed action includes owner, action, date, and verification.

FR-QLT-005 — Quality reporting

Requirement: Report inspection outcomes, rejection, open alerts, repeated issues, and corrective-action status.

Rationale: Quality leaders need measurable controls.

Preconditions: Metric definitions and time windows are documented.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Reports trace to inspections and alerts.

FR-QLT-006 — Supplier/product trends

Requirement: Support analysis by supplier, product, lot, and operation where records support it.

Rationale: Trend analysis helps prioritize corrective work.

Preconditions: Only comparable, defined measures are aggregated.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: A reported trend can be traced to included inspection records.

21. Studio and Configuration Governance

Purpose and scope

Allow approved adaptations of forms, fields, views, automation, and workflows while preserving access controls and maintainability.

Principal business records

Custom field, view, model, automated action, access group, configuration parameter, change request, release.

Functional requirements

FR-STD-001 — Customization authorization

Requirement: Restrict production-impacting customization to authorized administrators and approved change procedures.

Rationale: Uncontrolled customization alters core behavior.

Preconditions: Production changes require purpose, owner, and validation evidence.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Unauthorized user cannot change protected configuration.

FR-STD-002 — Field and view governance

Requirement: Document custom fields, types, defaults, dependencies, views, and access implications.

Rationale: Custom data must remain supportable.

Preconditions: Sensitive fields require explicit access/export rules.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Field appears only where intended and validates correctly.

FR-STD-003 — Automation safeguards

Requirement: Validate automated actions and prevent unintended repeated execution or loops.

Rationale: Automation can affect many records quickly.

Preconditions: High-impact automation is tested and assigned an owner.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: A test proves trigger behavior and safe failure logging.

FR-STD-004 — Change testing and release

Requirement: Test configuration changes outside production and document rollback for material changes.

Rationale: Changes may affect multiple modules.

Preconditions: Release approval and backup requirements follow policy.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Change record identifies test evidence, approver, release time, and rollback.

FR-STD-005 — Configuration inventory

Requirement: Maintain inventory of material customizations and integration settings with purpose and owner.

Rationale: Support teams need to understand deployed solution.

Preconditions: Secrets are not stored in plain-text business records.

Main flow: Open the relevant workspace; enter or select validated business data; review calculations and warnings; save or confirm the permitted transaction; verify linked history.

Exception handling: Missing mandatory values, invalid state transitions, insufficient permissions, duplicate submissions, or unavailable dependencies shall be reported without silently committing an invalid transaction.

Acceptance evidence: Reviewers can identify purpose and owner of material customization.

22. Business Rules

ID	Rule	Mandatory behavior
BR-001	Master data ownership	Each master-data domain shall have an assigned owner for definitions, duplicate handling, and lifecycle policy.
BR-002	Unique identifiers	Records requiring references shall use configured sequences or unique keys; duplicates are rejected with useful feedback.
BR-003	Draft versus final	Finalized records use controlled amendment, cancellation, reversal, or return workflows rather than silent overwrite.
BR-004	Approval separation	Where required, the initiating user shall not approve their own transaction unless an approved exception exists.
BR-005	Quantity consistency	Ordered, received, delivered, consumed, returned, invoiced, and credited quantities remain separately identifiable.
BR-006	Stock availability	On-hand, reserved, forecast, and available stock are distinct concepts.
BR-007	Financial posting	Only validated documents with required accounts, journals, tax settings, and dates may be posted.
BR-008	Currency and rounding	Exchange rates, precision, rounding, and exchange differences follow approved accounting configuration.
BR-009	Tax authority	Tax rates and fiscal mappings are maintained by authorized finance personnel and verified against jurisdictional policy.
BR-010	Effective dates	Contracts, subscriptions, rentals, employee changes, and date-sensitive records preserve effective start/end dates.
BR-011	Time-zone consistency	Stored timestamps and displayed local times follow a defined time-zone policy; date-only fields do not shift unexpectedly.
BR-012	Record linkage	Related records use stable identifiers and retain source links through completion, return, cancellation, and correction.
BR-013	Deletion control	Financial, stock, audit, legal, or operational records cannot be hard-deleted by ordinary users after finalization.
BR-014	Attachment security	Attachment permissions are no weaker than permissions for the protected record.
BR-015	Notification accuracy	Notifications identify record, action, owner, and due date without exposing restricted data.
BR-016	Error transparency	Validation failures identify corrective action; technical diagnostics remain restricted.
BR-017	Safe retry	Retrying an interrupted operation does not create duplicate orders, invoices, payments, receipts, or stock movements.
BR-018	Report reproducibility	Reports expose period, company, currency basis, and filters for reproducibility.
BR-019	Retention	Retention/deletion periods are configured to meet legal, contractual, and organizational requirements.
BR-020	Configuration change control	Changes to accounting, tax, access, stock, production, or automation settings require impact review and testing.

23. Data Requirements

23.1 Core master data

Entity	Minimum attributes	Validation / stewardship
Company	Name, legal identifier where applicable, address, currency, fiscal settings	Authorized administration; company boundaries enforced
Customer/contact	Name, contact channels, addresses, commercial terms, applicable tax IDs	Duplicate checks and mandatory fields
Supplier	Name, contacts, payment terms, currency, applicable tax details	Blocked supplier control and duplicate detection
Product/service	Name, reference, type, unit, sales/purchase, taxes, tracking settings	Unique reference and compatible stock/accounting setup
Warehouse/location	Code, type, parent, company, usage and route attributes	Valid hierarchy and operation scope
Employee	Employee reference, work contact, department, manager, employment state	Restricted access and lifecycle control
Account/journal	Code, name, type, company, currency/restrictions	Authorized finance maintenance and posting validation
Business document	Reference, company, owner, state, timestamps, linked references	Stable ID, valid transition, audit trail
Attachment/article	Title, type, owner, linked record, revision/publication, access group	Parent access parity, retention, publication control

23.2 Transaction data

Transactional records shall retain the parties, line items, quantities, prices, taxes, dates, state, responsible user, company, and linked source documents required by the relevant process. A confirmed sales order remains traceable to delivery and invoice records; a purchase order remains traceable to receipts and supplier bills; a manufacturing order remains traceable to component consumption and finished output.

23.3 Data quality

- Enforce mandatory fields before a record enters a state requiring them.
- Validate references against records in the permitted company and access scope.
- Use controlled values for status, priority, severity, and type where consistent reporting is required.
- Validate date order, amount precision, quantity precision, and unit conversion.
- Bulk imports provide row-level validation feedback and do not silently discard invalid rows.
- Duplicate detection allows users to distinguish true duplicates from legitimate similar records.

23.4 Retention and archive

The organization shall define retention periods for accounting entries, invoices, payments, employee records, support communications, attachments, and audit history. Archival preserves identifiers and relationships. Legal holds, active disputes, and statutory obligations override routine deletion. Exact durations are policy decisions.

23.5 Migration

Migration shall define entity dependency order, source-to-target mappings, transformations, duplicate handling, rejected-row disposition, reconciliation totals, owners, and sign-off. A sample and full-volume trial shall be completed before cutover. Opening balances and open transactions are reconciled by process owners.

23.6 Audit and provenance

For significant transactions, retain who created, changed, approved, posted, cancelled, or reversed the record and when. Linked documents retain origin references. Where a reason or supporting evidence is required, capture it before finalization.

24. External Interface Requirements

24.1 User interface

- Present navigation, list/form views, status indicators, actions, and validation messages consistently.
- Distinguish required and computed fields; confirm destructive or irreversible actions where appropriate.
- Preserve user-entered values after recoverable validation errors.
- Support keyboard navigation and readable contrast for common workflows.
- Provide relevant search, filter, sort, group, pagination, and export options subject to permission.

24.2 APIs and integrations

Enabled interfaces shall use documented endpoints, authentication, authorization, validation, error responses, and versioning. Transaction-creating requests shall use a stable external reference or idempotency key where supported. Integration logs shall retain correlation identifiers, outcome, time, and safe diagnostics without credentials or unnecessary personal data.

Interface	Expected behavior	Controls
Identity provider / SSO	Authenticate users and map identities to roles	Secure token handling, lifecycle, session expiry
Email / messaging	Send approved transactional and activity notifications	Delivery status, retry, privacy-aware content
Payment provider	Initiate payment and receive verified outcome	Provider reference, callback verification, idempotency
Shipping / carrier	Request shipment/label and receive tracking	Address validation, retry and exception queue
Banking / statement import	Import lines and support reconciliation	File validation, duplicate detection, audit totals
Payroll / statutory service	Exchange approved payroll/compliance data	Encryption, restricted scope, approval, acknowledgment
File import/export	Validate structured rows and report errors	Schema, encoding, row-level outcomes, permissions

24.3 Hardware interfaces

If barcode scanners, receipt printers, cash drawers, weighing devices, or other peripherals are used, the deployment shall document supported models, browser/device requirements, connector needs, failure behavior, and manual fallback. Peripheral failure must not silently duplicate an order or payment.

24.4 Error conventions

Interface errors distinguish validation, authorization, dependency outage, timeout, duplicate submission, and unexpected server failure. User messages remain actionable and avoid stack traces, database internals, secrets, or unauthorized personal information.

25. Non-Functional Requirements

ID	Quality attribute	Requirement
NFR-001	Performance	Under approved production load, 95% of ordinary page actions should complete within 3 seconds, excluding external-provider latency and designated large reports. Confirm by testing.
NFR-002	Transaction response	Normal transaction submissions should return success or actionable validation within 5 seconds under agreed load, excluding long jobs and external delays.
NFR-003	Scalability	Support forecast users, companies, products, transactions, attachments, and history without violating agreed performance targets.
NFR-004	Availability	Meet a business-approved availability target; define maintenance windows and dependency exclusions in the service plan.
NFR-005	Recoverability	Backups cover database, required attachments/filestore, configuration, and state needed for a consistent restore; restore tests are required.
NFR-006	Data integrity	Preserve referential integrity, prevent duplicate finalization on retries, and ensure atomicity for related changes.
NFR-007	Security	Enforce least privilege, server-side authorization, secure transport, protected secrets, session controls, and timely security updates.
NFR-008	Privacy	Expose personal and sensitive data only for approved purposes and authorized roles; apply retention/export policy.
NFR-009	Auditability	Trace material state changes, approvals, postings, access changes, and administrative actions according to policy.
NFR-010	Usability	Use consistent terminology, predictable navigation, clear state indicators, and field-level validation.
NFR-011	Accessibility	Target the organization-selected accessibility standard and verify keyboard use, focus, contrast, and labels.
NFR-012	Compatibility	Maintain a matrix of supported browsers, screen sizes, devices, peripherals, and Odoo release.
NFR-013	Maintainability	Document and assign owners to customizations/integrations; test and version material changes; maintain rollback plans.
NFR-014	Observability	Provide authorized logs, error trends, job status, integration outcomes, and health indicators for support.
NFR-015	Localization	Test currency, language, time zone, date/number formats, fiscal calendar, and tax configuration for each company.
NFR-016	Interoperability	Use documented schemas, stable identifiers, explicit error handling, and controlled version changes.
NFR-017	Capacity	Monitor database growth, attachments, logs, queues, and backup storage against thresholds.
NFR-018	Upgradeability	Use controlled upgrade process with compatibility review, test migration, data validation, and sign-off.
NFR-019	Confidentiality	Prevent restricted HR, finance, customer, and support data from appearing in unauthorized reports, exports, notifications, or attachments.
NFR-020	Resilience	External-service failure is visible and recoverable without silently losing or duplicating a business transaction.

25.1 Performance verification

Test representative data volumes, concurrent users, background jobs, and intended deployment settings. Define workload mix, duration, success criteria, and excluded latency. Report response-time percentiles, error rate, resource utilization, and tested build/configuration.

25.2 Recovery objectives

Business and operations owners shall define RTO, RPO, backup frequency, retention, maintenance windows, escalation contacts, and restoration responsibilities. A backup is verified only when a restore test demonstrates consistent recovery of the application and required attachments.

25.3 Security verification

Verify role/company boundaries, direct URL/API access, attachment authorization, session expiration, privileged configuration, secret handling, export permissions, and integration callbacks. Findings shall be classified, assigned, and resolved under the organization’s risk process.

26. Access Control and Separation of Duties

Capability	Operator	Manager	Finance	HR/payroll	Admin	Auditor
View operational records	Assigned	Team scope	Finance scope	HR scope	As required	Read-only scope
Create records	Scoped	Scoped	Finance scope	HR scope	As required	No
Approve	Delegated only	Within limits	Finance approvals	HR/payroll approvals	Avoid routine approvals	No
Post accounting	No by default	Only if assigned	Authorized role	Only if assigned	Emergency policy	No
View compensation	No	Policy-dependent	No by default	Authorized	Controlled support	Explicit authorization
Change access/config	No	No	No	No	Authorized admin	No
Export sensitive data	Restricted	Restricted	Finance scope	HR scope	Controlled	Approved read-only scope

This baseline must be mapped to actual Odoo groups, record rules, company rules, model permissions, and actions. Process owners and the security owner shall approve the final role matrix before production.

26.1 Privileged access

Administrative accounts shall be limited to named personnel, protected by the organization authentication policy, and used for business transactions only when necessary. Changes to permissions, financial settings, automation, or production configuration shall be logged and reviewed. Emergency access shall be time-limited and reviewed after use.

26.2 Separation of duties

Where feasible, creation, approval, posting, payment execution, and reconciliation shall be separated. If combined duties are unavoidable, the business owner shall approve compensating controls such as secondary review, exception reports, or periodic access review.

27. Reports, Dashboards and Monitoring

Area	Required report coverage
Sales	Quotation conversion, order book, delivery backlog, invoice status, sales by product/customer/team, returns and credits.
Purchasing	Open orders, overdue receipts, supplier variances, bills awaiting match, purchase commitments.
Inventory	On-hand/reserved/forecast, low stock, movements, lot/serial traceability, aging, adjustment variance.
Manufacturing	Production status, material shortages, planned versus actual output, work duration, scrap and quality holds.
Accounting	General ledger, trial balance, profit and loss, balance sheet, aged receivables/payables, reconciliation, tax summaries.
POS	Sales by session/cashier/payment method/product, returns, expected versus counted cash, session variance.
HR/payroll	Headcount, contract expiries, payroll totals and exceptions with restricted employee-level access.
Helpdesk	Ticket volume, aging, priority, team workload, service targets, reopened cases, resolution trends.
Projects	Task progress, overdue work, milestones, workload and approved timesheet summaries.
Maintenance	Open requests, preventive work due, overdue equipment, downtime and work history.
Subscriptions/rental	Upcoming renewals, billing exceptions, current rentals, overdue returns, utilization.
Quality	Inspection outcomes, nonconformance backlog, supplier/product trends, corrective-action status.
Documents/website	Publication status, document ownership, pending reviews, expired content where supported.

27.1 Report controls

- Every report shall expose its period, company, and active filters.
- Measures such as overdue, completed, utilization, aging, and conversion shall have documented definitions.
- Financial reports use posted accounting data and approved currency/fiscal rules unless clearly labeled otherwise.
- Exports respect row-level and field-level permissions.
- Sensitive HR/payroll and customer reports are restricted and auditable.

28. Workflow, Validation and Exception Handling

28.1 Validation principles

Validation shall occur at the earliest point where a rule can be checked and again on the server at submission. Messages shall explain corrective action. Business validation failures shall be distinguished from permission failures, external-service failures, duplicate requests, and unexpected technical errors.

Scenario	Required behavior
Missing mandatory field	Identify field and prevent state transition requiring it.
Insufficient permission	Reject action, avoid restricted disclosure, and record event where required.
Duplicate reference	Prevent duplicate finalization and provide safe route to inspect existing record if authorized.
Insufficient stock	Show shortage and permitted options; do not claim unavailable quantity was shipped.
Missing accounting configuration	Block posting and identify configuration category.
External service unavailable	Preserve recoverable state and support controlled retry without duplicate transaction.
Concurrency conflict	Detect stale/conflicting updates and request refresh or resolution.
Invalid lifecycle transition	Reject and explain current state or required prior step.
Invalid import row	Report row and reason; do not silently convert invalid values.
Background job failure	Record status and safe diagnostics; provide authorized retry/escalation.

28.2 Technical evidence

Diagnostics shall include timestamp, affected operation/module, correlation reference where available, severity, and sanitized exception information. Logs shall not contain passwords, tokens, payment credentials, or unnecessary personal data. Support users may connect an error to the relevant record without gaining broader data access.

28.3 Retry and recovery

Retryable operations distinguish not started, in progress, succeeded, failed, and outcome unknown. When a dependency times out and outcome is unknown, reconcile external reference or existing transaction before creating a replacement. Retries must be safe and observable.

29. System Acceptance Criteria

ID	Acceptance area	Pass condition
AC-001	Core access	Authentication, role/company scope, and denial of unauthorized screen/report/attachment/API access pass.
AC-002	Master data	Approved records can be created, validated, searched, updated, and archived by authorized roles.
AC-003	Order-to-cash	Quotation is approved, order confirmed, goods delivered, invoice issued, payment registered, and account reconciled with references preserved.
AC-004	Procure-to-pay	Purchase order approved, goods received, supplier bill matched, payment recorded, and payable reconciled.
AC-005	Inventory integrity	Receipt, transfer, reservation, delivery, return, count, and lot/serial scenarios produce expected balances/history.
AC-006	Manufacturing	Order consumes components, records output, handles quality result, and closes with traceable variance.
AC-007	Accounting	Invoice, bill, payment, tax, posting, reversal, period lock, and report scenarios meet approved expected outcomes.
AC-008	POS	Sale, payment, receipt, refund, session close, and reconciliation are traceable by payment method.
AC-009	HR/payroll	Personnel data is restricted and payroll follows approved input, review, confirmation, and accounting process.
AC-010	Service workflows	Tickets, tasks, maintenance requests, and quality alerts can be assigned, tracked, closed, and reported with history.
AC-011	Recurring/rental	Renewal billing and rental availability/pickup/return/charges follow approved dates, prices, and lifecycle rules.
AC-012	Website/eCommerce	Published content/catalog is correct; online order creates one traceable sale and payment outcome.
AC-013	Document security	Private documents and internal notes are inaccessible to unauthorized users via links, exports, or related screens.
AC-014	Migration	Reconciled counts/totals meet signed tolerances and rejected records have documented disposition.
AC-015	Performance	Representative load tests meet approved response, throughput, and error-rate targets.
AC-016	Recovery	Restore test recovers application, database, attachments, and configuration within approved objectives.
AC-017	Auditability	Reviewer can trace creation, material changes, approval, finalization, and correction of representative transactions.
AC-018	Reporting	Key operational and financial reports reconcile to records for agreed periods and filters.

29.1 Acceptance process

Use approved scenarios and representative data. Mark each result pass, fail, blocked, or not applicable and attach evidence/defect references. A requirement is accepted only when its expected result is demonstrated and any deviation is corrected or formally accepted by the accountable owner.

29.2 Exit criteria

- All critical business scenarios pass in the release candidate.
- No unresolved defect violates mandatory financial, access, data-integrity, or stock-integrity requirements unless formally accepted.
- Migration and opening balances meet signed reconciliation tolerances.
- Runbooks, backup/restore evidence, role matrix, integration contacts, and rollback instructions are approved.
- Process owners sign off on localization, approval thresholds, and report definitions.

30. Traceability and Change Management

30.1 Traceability

Each requirement shall link to a business objective/process, a configuration or design decision, and one or more verification cases. Maintain a traceability register so requirement status can be updated as the solution changes.

Requirement family	Primary process	Verification evidence
FR-COM	Cross-module administration	Role, lifecycle, audit, duplicate and retry tests
CRM / Sales	Lead-to-cash	Qualification, quotation approval, fulfillment, invoicing
Purchase / Inventory	Procure-to-pay and stock	Approval, receipt match, transfer, stock reconciliation
Manufacturing / Quality	Plan-to-produce	BoM, consumption, quality gate, production close
Accounting	Record-to-report	Posting, payment, reconciliation, period lock, reports
HR / Payroll	Hire-to-pay	Restricted access, contracts, payroll and accounting
Helpdesk / Project / Maintenance	Service delivery	Assignment, due dates, history, reporting
Subscriptions / Rental	Recurring service	Renewal, billing, reservation, return, charges
Website / Documents / Studio	Content and administration	Publication, permissions, customization approval
NFR	Cross-cutting quality	Performance, security, recovery, compatibility, resilience

30.2 Change request contents

A change request shall state requirement identifiers, business reason, affected owners/modules/integrations, migration impact, security/privacy impact, testing scope, approval, release target, and rollback/contingency. Changes to tax, payroll, stock valuation, permissions, accounting, or automation require explicit review by the accountable owner.

30.3 Baseline policy

Maintain one approved SRS version. Draft changes are not approved production behavior. Retain identifiers when intent remains substantially the same; issue new identifiers when a requirement becomes materially different or splits into independently testable behavior.

31. Assumptions, Open Decisions and Risks

31.1 Decisions required before configuration

- Confirm Odoo version, edition, installed modules, localization, hosting, and supported browsers.
- Approve company structure, multi-company boundaries, warehouses, stock routes, valuation method, and units of measure.
- Approve chart of accounts, journals, tax rules, fiscal periods, payment methods, invoice policies, and period locks.
- Define sales/purchase approval thresholds, discount limits, credit policy, refund rules, and separation of duties.
- Define HR privacy boundaries, employment states, payroll localization, compensation rules, and reports.
- Define subscription billing/renewal/cancellation policy and rental grace periods, damage charges, deposits, and return inspections.
- Define helpdesk categories, priorities, service calendars, response/resolution targets, escalation, and customer communication.

- Define retention periods, audit requirements, legal holds, publication roles, and export controls.
- Identify each integration owner, data contract, authentication, retry behavior, service target, and support contact.
- Set measurable performance, availability, recovery, accessibility, and support targets.

31.2 Key risks

Risk	Potential effect	Mitigation
Unclear master-data ownership	Duplicate/inconsistent records	Assign stewards, validate, clean duplicates, reconcile migration
Unapproved financial configuration	Incorrect tax/accounting/valuation	Finance sign-off and scenario-based posting tests
Excessive permissions	Unauthorized data or transactions	Least privilege, role matrix, access testing, periodic review
Uncontrolled customization	Upgrade difficulty and inconsistent workflows	Change control, ownership, test environment, regression suite
Unsafe retry behavior	Duplicate order/payment/stock movement	Stable references, idempotency, reconcile before retry
Incomplete migration	Missing balances or open transactions	Dependency-ordered migration, trial load, reconciliation
Unclear process ownership	Exceptions remain unresolved	RACI, queues, escalation, acceptance owners
Inadequate recovery test	Long outage or incomplete restore	RTO/RPO, full restore test, runbooks

31.3 Assumptions

The organization will appoint process owners, approve configuration choices, supply valid master data, and provide representative acceptance scenarios. Standard Odoo functionality is preferred where appropriate; exact module availability and behavior must be confirmed against the selected release. Configurable rules are not final until an authorized owner records a decision.

32. Glossary

Term	Definition
Accounting period	Defined period used for accounting reporting and posting control.
Approval workflow	Controlled sequence of reviews/authorizations before a transaction proceeds.
BoM	Bill of Materials: components and quantities required to manufacture a product.
Company boundary	Data and transaction scope associated with a legal or operational company.
Credit note	Financial document reducing or correcting a previously issued invoice.
Idempotency	Retrying the same request does not create a duplicate business effect.
Lot/serial	Identifier used to trace a batch or individual item through stock and production.
Master data	Reference information such as products, customers, suppliers, employees, accounts, and locations.
Nonconformance	Recorded failure to meet an approved requirement or quality criterion.
Posted document	Finalized document that has affected accounting or another controlled ledger.
Reconciliation	Matching records/balances and resolving differences.
Reservation	Allocation of available inventory to a specific demand or operation.
Role-based access control	Authorization based on assigned roles and permitted scope.
SLA	Service-level agreement defining expected response/resolution target.
Stock move	Record representing quantity movement between stock locations or related stock operation.
Traceability	Ability to follow a record through origin, processing, correction, and outcome.

Appendix A. Suggested End-to-End Acceptance Scenarios

ID	Scenario	Steps / expected outcome
A-01	Sales order with available stock	Create quotation → approve/confirm → reserve → deliver → invoice → payment → reconcile.
A-02	Sales order with shortage	Confirm order → show shortage/backorder → receive replenishment → fulfill remainder.
A-03	Purchase with partial receipt	Approve PO → partial receipt → record discrepancy → receive remainder → match bill.
A-04	Supplier bill mismatch	Register bill with variance → flag → approve/correct → post only after resolution.
A-05	Lot/serial traceability	Receive tracked item → transfer/consume → deliver/produce → trace origin and destination.
A-06	Manufacturing quality failure	Create order → fail inspection → hold/disposition → corrective action → approved completion.
A-07	Financial period lock	Unauthorized backdated posting is rejected; allowed override is approved and logged.
A-08	POS refund	Complete sale → locate original → refund permitted lines → reconcile payment and stock.
A-09	Payroll privacy	General user is denied employee-level payroll access; authorized user reviews batch.
A-10	Support escalation	Create ticket → assign → breach service target → escalate → resolve → reopen and inspect history.
A-11	Rental conflict	Create reservation → attempt overlapping reservation → block or route for approved resolution.
A-12	Subscription renewal	Create recurring contract → generate scheduled invoice → renew/cancel → verify linked billing.
A-13	Document access	Upload restricted file → test role access through UI, direct link, and export.
A-14	Integration retry	Submit transaction → simulate timeout → retry same reference → verify one business effect.
A-15	Restore and reconcile	Restore backup in test environment → verify database, attachments, users, config, and totals.

Appendix B. Requirements Coverage Index

Area	Requirement identifiers
Common services	FR-COM-001 to FR-COM-015
CRM	FR-CRM-001 to FR-CRM-006
Sales	FR-SAL-001 to FR-SAL-006
Purchase	FR-PUR-001 to FR-PUR-006
Inventory	FR-INV-001 to FR-INV-006
Manufacturing	FR-MFG-001 to FR-MFG-006
Accounting	FR-ACC-001 to FR-ACC-006
Point of Sale	FR-POS-001 to FR-POS-006
HR and Payroll	FR-HR-001 to FR-HR-003; FR-PAY-001 to FR-PAY-003
Helpdesk	FR-HD-001 to FR-HD-006
Project	FR-PRJ-001 to FR-PRJ-005
Maintenance	FR-MNT-001 to FR-MNT-006
Subscriptions	FR-SUB-001 to FR-SUB-006
Rental	FR-RNT-001 to FR-RNT-006
Website/eCommerce	FR-WEB-001 to FR-WEB-006
Documents/Knowledge	FR-DOC-001 to FR-DOC-006
Studio/configuration	FR-STD-001 to FR-STD-005
Non-functional	NFR-001 to NFR-020
Business rules	BR-001 to BR-020
Acceptance criteria	AC-001 to AC-018

Appendix C. Review Checklist

- Process owners have reviewed module scope and requirements relevant to their departments.
- Configuration-dependent values have a named owner and approved decision.
- Access control, separation of duties, privacy, and export requirements are reviewed.
- Financial and inventory workflows have test cases with expected accounting/quantity outcomes.
- Integration contracts and retry/error behavior are documented for selected services.
- Migration, retention, backup, restore, and support procedures have assigned owners.
- Acceptance evidence is linked to requirement identifiers and approved release baseline.

End of Software Requirements Specification